

## DISKO 3

⚠ **Security Warning:** Work in an isolated environment (virtual machine or container). Never open unknown files on your primary machine.

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### Challenge Description

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Can you find the flag in this disk image? This time, it's not as plain as you think it is!

### Hints

- How will you search and extract files in a partition?
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### Tools Used

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- `7z` – Extract compressed files
  - `file` – Determine file type
  - `strings` – Extract printable character sequences
  - `exiftool` – Read metadata
  - `fdisk` – View partition tables
  - `mount` – Mount a disk image as a loopback device
  - `ls` – List directory contents
  - `cp` – Copy files
  - `gunzip` – Decompress `.gz` files
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### Complete Solution

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#### Step 1: Download and Extract

Download the disk image named `disko-3.dd.gz` from the challenge page.

Extract the compressed disk image:

```
7z x disko-3.dd.gz
```

This decompresses the gzipped file and creates the raw disk image `disko-3.dd`.

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## Step 2: Basic File Inspection

Verify the file type and understand its structure:

```
file disko-3.dd
```

### Expected output:

```
disko-3.dd: DOS/MBR boot sector, code offset 0x58+2, OEM-ID "mkfs.fat", Media descriptor 0xf8, sectors/track 32, heads 8, sectors 204800 (volumes > 32 MB), FAT (32 bit), sectors/FAT 1576, reserved 0x1, serial number 0x49838d0b, unlabeled
```

The output confirms this is a **FAT32** filesystem. This is commonly used on USB drives and memory cards.

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## Step 3: Search for Visible Text

Disk images often contain strings embedded directly in the data. Extract all readable character sequences and search for the flag:

```
strings disko-3.dd | grep -i "picoCTF"
```

### Output (excerpt):

```
MESSAGE=[system] Activating via systemd: service name='org.bluez' unit='dbus-org.bluez.service' requested by ':1.81' (uid=1000 pid=12105 comm="/usr/share/code/code Desktop/picoctf-2025")
MESSAGE=[system] Activating via systemd: service name='org.bluez' unit='dbus-org.bluez.service' requested by ':1.66' (uid=1000 pid=43129 comm="/usr/share/code/code Desktop/picoctf-2025")
MESSAGE=[system] Activating via systemd: service name='org.bluez' unit='dbus-org.bluez.service' requested by ':1.65' (uid=1000 pid=2141 comm="/usr/share/code/code
```

```
Desktop/picoctf-2025")
MESSAGE=[system] Activating via systemd: service name='org.bluez' unit='dbus-
org.bluez.service' requested by ':1.65' (uid=1000 pid=2584 comm="/usr/share/code/code
Desktop/picoctf-2025")
```

No flag appears yet. We need to dig deeper.

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## Step 4: Inspect Metadata

Examine the disk image metadata:

```
exiftool disko-3.dd
```

### Expected output:

|                             |                             |
|-----------------------------|-----------------------------|
| ExifTool Version Number     | : 13.50                     |
| File Name                   | : disko-3.dd                |
| Directory                   | : .                         |
| File Size                   | : 105 MB                    |
| File Modification Date/Time | : 2026:05:09 14:37:36+02:00 |
| File Access Date/Time       | : 2026:05:09 14:51:24+02:00 |
| File Inode Change Date/Time | : 2026:05:09 14:37:36+02:00 |
| File Permissions            | : -rw-r--r--                |
| Error                       | : Unknown file type         |

Nothing suspicious here.

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## Step 5: View Partitions

Use `fdisk` to see the partition structure:

```
fdisk -l disko-3.dd
```

### Expected output:

```
Disk disko-3.dd: 100 MiB, 104857600 bytes, 204800 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
```

I/O size (minimum/optimal): 512 bytes / 512 bytes

Disklabel type: dos

Disk identifier: 0x00000000

There are **no explicit partitions**, which hints that a filesystem is written directly to the image (a “superfloppy” or raw filesystem image).

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## Step 6: Mount the Disk Image

Since the image contains a FAT32 filesystem directly (no partition table), we can mount it as a loopback device:

```
sudo mkdir /mnt/disko-3
```

```
sudo mount -o loop disko-3.dd /mnt/disko-3
```

**Note:** The `-o loop` option allows Linux to treat a regular file as a block device.

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## Step 7: Explore the Mounted Image

After mounting, list its contents:

```
ls /mnt/disko-3
```

### Output:

```
log
```

Let's check inside the `log/` directory:

```
ls -l /mnt/disko-3/log
```

### Output:

```
total 1608
-rwxr-xr-x 1 root root 21322 Mar 31 2025 Xorg.0.log
-rwxr-xr-x 1 root root 20753 Mar 31 2025 Xorg.0.log.old
```

```
-rwxr-xr-x 1 root root      0 Mar 31  2025 alternatives.log
-rwxr-xr-x 1 root root    194 Mar 31  2025 alternatives.log.2.gz
drwxr-xr-x 2 root root   1536 Mar 31  2025 apt
-rwxr-xr-x 1 root root      0 Mar 31  2025 boot.log
-rwxr-xr-x 1 root root   7563 Mar 31  2025 boot.log.1
-rwxr-xr-x 1 root root   7125 Mar 31  2025 boot.log.5
-rwxr-xr-x 1 root root   7420 Mar 31  2025 boot.log.6
-rwxr-xr-x 1 root root 299509 Mar 31  2025 daemon.log
-rwxr-xr-x 1 root root  24374 Mar 31  2025 debug
-rwxr-xr-x 1 root root   3378 Mar 31  2025 dpkg.log.1
-rwxr-xr-x 1 root root      0 Mar 31  2025 dpkg.log.2.gz
-rwxr-xr-x 1 root root 161557 Mar 31  2025 dpkg.log.4.gz
-rwxr-xr-x 1 root root 259660 Mar 31  2025 dpkg.log.5.gz
-rwxr-xr-x 1 root root  32032 Mar 31  2025 faillog
-rwxr-xr-x 1 root root     78 Jul 17  2025 flag.gz
drwxr-xr-x 3 root root    512 Mar 31  2025 inetsim
drwxr-xr-x 3 root root   1024 Mar 31  2025 installer
....
```

Notice the suspicious file: **flag.gz** (only 78 bytes). This stands out among the larger log files.

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## Step 8: Extract the Flag File

Copy the suspicious file to your current directory:

```
cp /mnt/disko-3/log/flag.gz .
```

Decompress it:

```
gunzip flag.gz
```

Read the extracted flag file:

```
cat flag
```

## Output:

```
Here is your flag
picoCTF{<REDACTED>}
```

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## Final Result

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The flag is obtained from the hidden `flag.gz` file inside the `log` directory:

```
picoCTF{<REDACTED>}
```

**Note:** The actual flag is redacted here. In the real challenge, the exact value will be displayed.

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## Key Concepts

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- **Raw Disk Image:** A sector-by-sector copy of a storage device, which can contain a filesystem without a partition table.
- **Loopback Mount:** Mounting a file as a block device using `mount -o loop`.
- **FAT32 Filesystem:** A simple, widely supported filesystem often used on removable media.
- **Steganography / Hiding Files:** The flag was hidden in a compressed `.gz` file named `flag.gz` inside a `log` directory, blending in with system logs.
- **Forensic Analysis:** Mounting and exploring the filesystem is a key technique for extracting hidden data from disk images.